

MHF 4U Chapter 2 Polynomial Equations and Inequalities Test

Name: _____

Date: Monday, October 7, 2024

Total

26

/29

KNOWLEDGE

1. What is the remainder when $4x^3 - 10x^2 + 6x - 18$ is divided by $2x - 5$?

a) -14

b) -158

(c) -3

d) -798

2. Which of the following does not belong to the same family?

(a) $y = x(2x + 3)(1 + 2x)(x - 2)$

b) $y = x(2x + 3)(2x - 1)(x - 2)$

c) $y = -2x(2x + 3)(4x - 2)(x - 2)$

d) $y = 3x(2x + 3)(1 - 2x)(2 - x)$

3. Which is not a factor of $6x^3 - 7x^2 + 1$?

a) $(x - 1)$

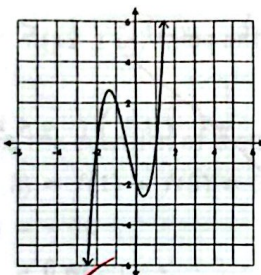
(b) $(6x + 1)$

c) $(3x + 1)$

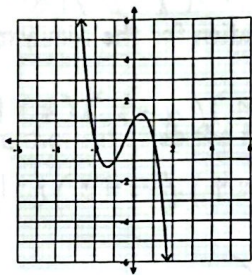
d) $(2x - 1)$

4. Which graph does not belong to the same family?

a)

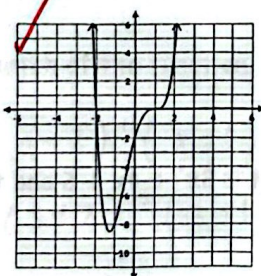


b)

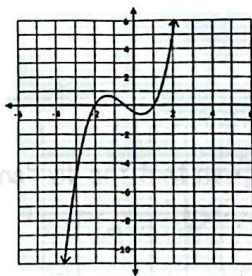


(5)

(c)



d)



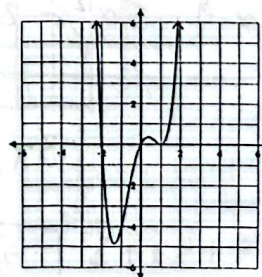
5. Given the graph of $f(x)$, when is $f(x) > 0$?

a) $-2 < x < 0$

b) $x < -2, x > 1$

c) $0 < x < 1$

(d) $x < -2, 0 < x < 1, x > 1$



6. Solve $x^4 + 2x^3 + 6x^2 + 18x = 27$. List all real and complex numbers (if any exist) in simplest form.

$$x^4 + 2x^3 + 6x^2 + 18x - 27 = 0$$

$$f(1) = 0 \rightarrow (x-1)$$

$$\begin{array}{r|rrrrr} 1 & 1 & 2 & 6 & 18 & -27 \\ + & & 1 & 3 & 9 & 27 \\ \hline -3 & 1 & 3 & 9 & 27 & 0 \end{array} \leftarrow f(-3) = 0 \rightarrow (x+3)$$

$$x^2 + 9 = 0$$

$$\sqrt{x^2} = \sqrt{-9}$$

$$x = \pm 3i$$

$$x = (1, 0), (-3, 0), 2.5$$

$$+ 3i, -3i$$

7. The zeros of a quartic function are $-4, -3, 1$ and 6 .

a) Determine an equation for the family of quartic functions with these zeros.

$$x = -4, -3, 1, 6$$

$$f(x) = k(x+4)(x+3)(x-1)(x-6)$$

b) Determine an equation for the family member whose graph passes through the point $(-1, 21)$.

$$f(x) = k(x+4)(x+3)(x-1)(x-6)$$

$$21 = k(-1+4)(-1+3)(-1-1)(-1-6)$$

$$\frac{21}{84} = \frac{84k}{84}$$

$$k = \frac{1}{4}$$

Application

1. Determine the divisor given that the dividend is $3x^3 - 5x^2 - 2x - 1$, the quotient is $3x^2 + 4x + 5$ and the remainder is $5x + 14$.

$$3x^3 - 5x^2 - 2x - 1 = \text{divisor} (3x^2 + 4x + 5) + 5x + 14$$

$$3x^3 - 5x^2 - 2x - 1 = (\text{divisor}) (3x^2 + 4x + 5) + 5x + 14$$

$$3x^3 - 5x^2 - 2x - 1 - 5x - 14 = \text{divisor}$$

$$3x^2 + 4x + 5 \rightarrow \text{should be } - (x-1)$$

$$\begin{array}{r|l} 3x^2 + 4x + 5 & 3x^3 - 5x^2 - 7x - 15 \\ - (3x^3 - 4x^2 + 5x - 15) & \\ \hline & 9x^2 + 12x - 15 \\ & 9x^2 + 12x - 15 \end{array}$$

the divisor is $(x+3)$

2. (a) Graph the function $f(x) = 2x^4 - 3x^3 - 9x^2 + 8x + 12$. Label the x- and y-intercepts.

$$f(-1) = 0 \rightarrow (x+1)$$

$$\begin{array}{r|rrrrr} -1 & 2 & -3 & -9 & 8 & 12 \\ & \downarrow & & & & \\ + & 2 & -5 & -9 & 12 & 0 \end{array}$$

$$f(2) = 0 \rightarrow (x-2)$$

$$\begin{array}{r|rrrr} + & 2 & -5 & -9 & 12 & 0 \\ & \downarrow & & & & \\ x & 2 & -1 & -6 & 0 \end{array}$$

$$\begin{array}{r|rrrr} + & 2 & -1 & -6 & 0 \\ & \downarrow & & & \\ x & 2 & -1 & -6 & 0 \end{array}$$

$$(2x+3)(x-2)$$

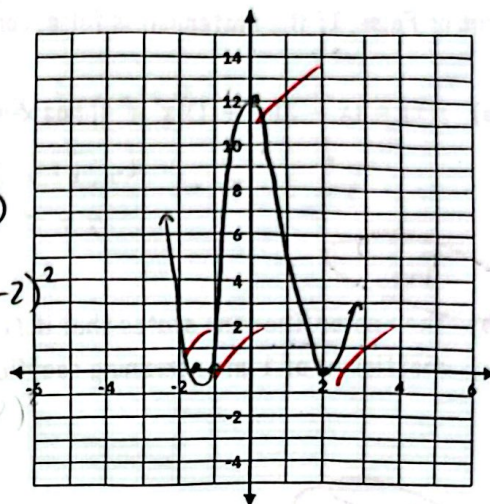
$$x = -\frac{3}{2}, 2, -1$$

(b) State the interval(s) when $f(x) > 0$.

$$f(x) > 0 \text{ when } (-\infty, -\frac{3}{2}) \cup (-1, 2) \cup (2, \infty)$$

$$f(x) = (x+1)(x-2)^2(2x+3)$$

$$y\text{-int: } 12$$



$$x\text{-int: } -\frac{3}{2}, 2, -1$$

$$y\text{-int: } 12$$

3. Determine the equation of a quartic function $f(x)$, in factored form, that satisfies the following conditions:

- $f(-2) = f(1) = f(5) = 0$
- $f(3) = -10$
- $f(x) > 0$ when $x < -2$, $-2 < x < 1$ or $x > 5$
- $f(x) < 0$ when $1 < x < 5$

Provide a sketch of the function.

$$f(x) = a(x+2)^2(x-1)(x-5)$$

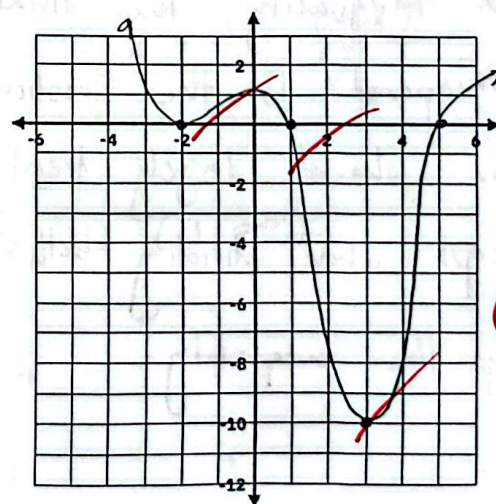
$$-10 = a(3+2)^2(3-1)(3-5)$$

$$-10 = a(-100)$$

$$a = \frac{1}{10}$$

$$a = \frac{1}{10}$$

$$f(x) = \frac{1}{10}(x+2)^2(x-1)(x-5)$$



(3)

Communication

1. True or False. If the statement is false, correct it.

a) $f(x) = (x - 3)(x + 1)(x^2 + 4)$ has x-intercepts at $x = -1, 3, -2i, 2i$

$\downarrow \quad \downarrow \quad \hookrightarrow \sqrt{x^2} = \sqrt{-4}$
 $x = 3 \quad x = -1 \quad x = \pm 2i$

imaginary # can't be x-int

True

b) The Factor Theorem states that if $(x - b)$ is a factor of the polynomial function $f(x)$ with a leading coefficient of 1 and remaining coefficients that are integers, then b is a factor of the constant term.

True

c) Polynomial functions within the same family have the same x- and y-intercepts.

False, polynomial functions with families must have the same x-intercepts and direction of opening, not y-intercepts.

2. What is the difference between solving a polynomial equation and a polynomial inequality?

In order to solve a polynomial equation, the final answer given is typically the x-intercepts. However, when solving an inequality, the final answer given are intervals that correspond to the question. When solving a polynomial inequality, you should largely treat the inequality sign like an $=$ sign, except when dividing both sides by a $(-)$ #, in which you flip the inequality.

ex.

answer for polynomial

$x = 1, -3, 12$

answer for inequality

$f(x) \geq$ when $[-3, 8]$

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thinking

HF4

1. If $f(x) = 2x^4 + 5x^3 - 14x^2 + 1$ and $g(x) = 2x^3 - 5x^2 - x - 2$, determine the values of x such that $f(x) > g(x)$ using an interval chart.

$$f(x) = 2x^4 + 5x^3 - 14x^2 + 1 \quad g(x) = 2x^3 - 5x^2 - x - 2$$

$$f(x) > g(x)$$

$$2x^4 + 5x^3 - 14x^2 + 1 > 2x^3 - 5x^2 - x - 2$$

$$2x^4 + 3x^3 - 9x^2 + x + 3 > 0$$

$$f(1) = 0 \rightarrow (x-1)$$

1		2	3	-9	1	3	
		↓					
+		2	5	-9	-3		
x		2	5	-9	-3	0	$f(1) = 0 \rightarrow (x-1)$
		↓					
+		2	7	3			
x		2	7	3	0		
		↓					
		2	1				
		↓					
		2	3				

$$(2x+1)(x+3)(x-1)^2$$

$$x = -\frac{1}{2}, -3, 1$$

interval
factor

$$(-\infty, -3) \quad (-3, -\frac{1}{2}) \quad (-\frac{1}{2}, 1) \quad (1, \infty)$$

$(2x+1)$	-	-	+	+
$(x+3)$	-	+	+	+
$(x-1)^2$	+	+	+	+
all	+	-	+	+

$$f(x) > g(x) \text{ when } (-\infty, -3) \cup (-\frac{1}{2}, 1) \cup (1, \infty)$$

(4)

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